Marco Martinez

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EXPERIENCE

Senior Software Engineer, FUJIFILM Healthcare Americas Corporation

Research & Development — Aug. 2026 - Present

- Built a centralized observability platform replacing fragmented file logging with container-native logging, metrics, and distributed tracing across .NET services.
- Developed reusable telemetry instrumentation and deployed telemetry collection infrastructure to Kubernetes using OpenTelemetry, Loki, Tempo, Prometheus, and S3-compatible storage.

Product Development — Mar. 2025 - Aug. 2026

- Delivered a full-stack third-party application integration using C#, .NET, TypeScript, and Razor Pages, including customer-specific deployment scripting, protected backups, and recovery safeguards, contributing to a successful $10M customer sale.
- Designed and rolled out a shared TypeScript/Razor Pages UI library across 5+ web applications, standardizing dialog behavior and driving cross-team adoption of a common implementation.
- Redesigned a SignalR-based enterprise chat application to use participant-based conversation identity, eliminating duplicate conversations and improving server-side query efficiency.
- Optimized PL/SQL chat queries by moving filtering and sorting into the database and limiting result sets, reducing response times from ~6 seconds to near-instant.
- Diagnosed and fixed a long-standing JavaScript memory leak caused by Chrome extension interactions that bypassed normal QA coverage, resolving a customer-impacting production issue.

Software Engineer II, LexisNexis — May 2023 - Mar. 2025

- Designed the T-SQL authentication data model and implemented schema migrations with FluentMigrator, securely storing API credentials using unique salts and cryptographic hashes.
- Designed and built a C#/.NET OData authentication flow, allowing JWT-authenticated users to provision API keys for Power BI access to on-premises reporting data.
- Served as the primary technical point of contact for authentication integrations, guiding application and automation teams through API-key and token-based authentication flows.
- Led offshore engineers implementing OData/.NET endpoints, assigning work around repeatable application patterns and establishing implementation conventions.
- Built a GraphQL telemetry framework using schema-level annotations to log request properties in the cloud, with configurable PII obfuscation to improve post-deployment troubleshooting.

Software Engineer I, LexisNexis — Feb. 2022 - May 2023

- Migrated multiple C#/.NET applications from implicit-flow authentication to PKCE.
- Implemented custom C#/.NET CORS and host-filtering middleware to secure applications in load-balanced cloud environments.
- Built and supported Angular/TypeScript application workflows backed by OData endpoints and optimized GraphQL data retrieval using data loaders.

Systems Engineer, Infosys — Nov. 2021 - Feb. 2022

Client: Walmart

- Fixed bugs across enterprise applications using Java, Spring Boot, Kubernetes, and PL/SQL.
- Applied object-oriented design patterns and established application conventions while implementing Spring Boot services.

PROJECTS

Yavria — AI Context & Knowledge Platform

C#, .NET 10, PostgreSQL, pgvector, React/TypeScript, Docker, OpenAI, OpenTelemetry

- Built a self-hosted, multi-service AI platform for persistent conversation context, semantic knowledge retrieval, file-based context, and external application integrations.
- Designed a retrieval architecture using PostgreSQL/pgvector, provider-agnostic embeddings, streaming LLM responses, and explicit service boundaries between conversation orchestration and knowledge storage.
- Containerized and deployed the platform with ingress routing through Cloudflare Tunnel, centralized authentication and API access, database migrations, and diagnostics.

NextProblem — Adaptive Coding Interview Practice Platform

C#, .NET 10, ASP.NET Core, React/TypeScript, OpenAI

- Built a full-stack coding-practice platform that tracks problem history and generates personalized daily practice queues based on mastery, completion state, and spaced repetition.
- Designed a focus-first planning system that concentrates practice on individual problem-solving patterns while periodically resurfacing previously learned material.
- Integrated Yavria as an external AI coaching service, supplying practice history and progress as context for personalized guidance while keeping the two systems independently deployable.

EDUCATION

B.S. Computer Science | Southern New Hampshire University